Decoding Parameter and Sampling Analysis

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1. Objective

This project examines how Temperature, Top-K and Top-P affect responses from two locally run language models. The comparison focuses on repeated wording, factual accuracy on three short questions, and compliance with a requested JSON structure.

2. Models and tools

Llama 3.2:3B (llama3.2:3b) and Qwen 2.5:3B (qwen2.5:3b) were run through Ollama. Python automated the requests. The Ollama library communicated with the models, Pandas stored and grouped the results, NumPy calculated averages, and Matplotlib produced the comparison graphs. The experiment was completed in a Jupyter notebook, with responses saved as CSV files.

3. Expected results

Before examining the results, lower sampling values were expected to produce more repeatable answers. Higher values were expected to allow more variation in wording. Factual accuracy and structure were compared separately because a different response is not necessarily an incorrect one. These expectations were treated as hypotheses rather than assumed outcomes.

4. Main finding

Both models answered the three short factual questions correctly and passed the automated JSON structure check in all 384 responses. Temperature changed how often the complete response was repeated exactly. Across all settings, Llama had higher exact agreement than Qwen, although neither model was more accurate on the questions that were scored.

5. Experimental method

Setting	Values
Temperature	0.0, 0.3, 0.7, 1.0
Top-K	10, 20, 50, 100
Top-P	0.5, 0.7, 0.9, 1.0
Seeds	42, 123, 456
Output limit / context	200 tokens / 2,048 tokens

Every combination of the three sampling parameters was tested: 4 x 4 x 4 = 64 settings. Three seeds were used for each setting and each model, giving 64 x 3 x 2 = 384 responses. The same prompt, seeds and limits were used for both models. Each request started with the user prompt only, without previous conversation history. JSON output was requested in the prompt but was not enforced through the API.

Fixed prompt

The prompt requested one JSON object with exactly four string fields: capital, planet, symbol and explanation. Markdown, code fences and text outside the object were prohibited. The questions were:

capital: What is the capital of Pakistan?

planet: Which planet is known as the Red Planet?

symbol: What is the chemical symbol for gold?

explanation: In one sentence, explain why Earth has seasons.

How the responses were scored

Factual accuracy: the proportion of three answers matching Islamabad, Mars and Au. Capital and planet answers were stripped of surrounding whitespace and compared without case sensitivity; Au was case-sensitive. Factual accuracy was evaluated only for the three short-answer questions; the seasons explanation was not assessed.

Structural compliance: a score of 1 required a valid JSON object containing exactly the four requested keys, with string values. Otherwise the score was 0. This check did not test whether the explanation was exactly one sentence.

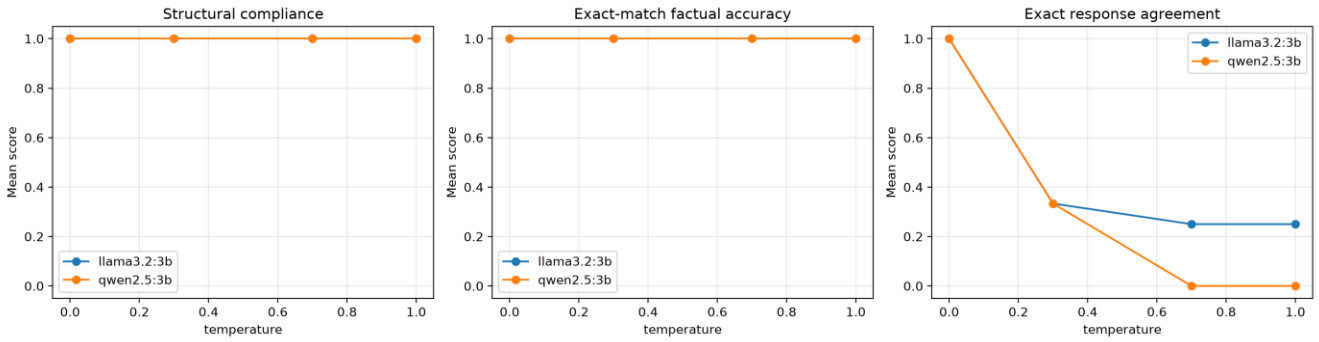
Exact agreement: the three responses at each setting formed three pairs. Agreement was the number of identical pairs divided by three, after stripping surrounding whitespace. This measures text consistency across seeds, including formatting differences; it does not measure similarity of meaning.

6. Results overview

The results contain 192 responses per model and no duplicate model/settings/seed entries. All responses ended with the recorded stop reason rather than the output-length limit. The summary contains 128 model/settings rows, each based on three responses.

Measure	Llama 3.2:3B	Qwen 2.5:3B
Responses	192	192
Mean short-answer accuracy	100%	100%
JSON structural compliance	100%	100%
Mean exact agreement	45.83%	33.33%

Temperature



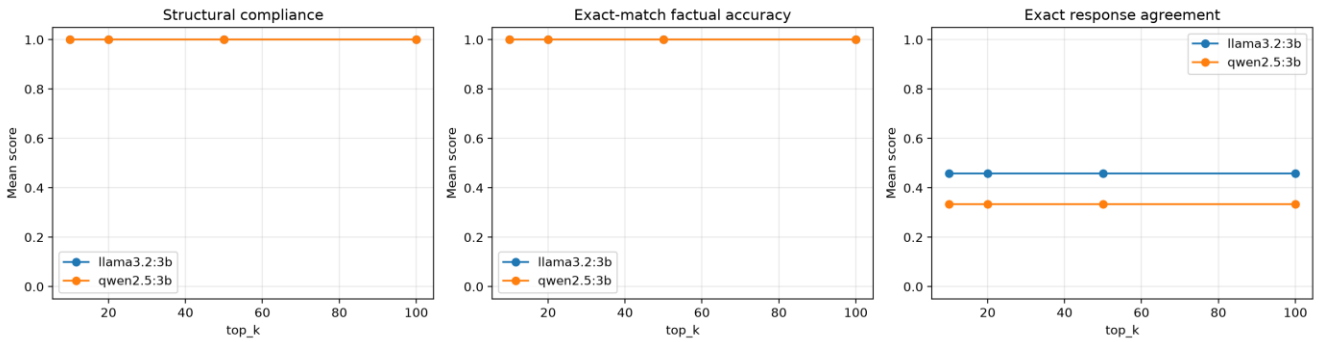
This Fig. shows mean scores by temperature. Each point averages over all Top-K and Top-P settings.

Temperature	Llama agreement	Qwen agreement
0.0	100%	100%
0.3	33.33%	33.33%
0.7	25%	0%
1.0	25%	0%

At temperature 0.0, each model repeated the same complete response across the three seeds within every parameter combination. Agreement fell at higher temperatures. At 0.7 and 1.0, Qwen produced three distinct responses within each combination, while Llama retained some identical pairs. The three scored facts and the checked JSON structure remained correct throughout.

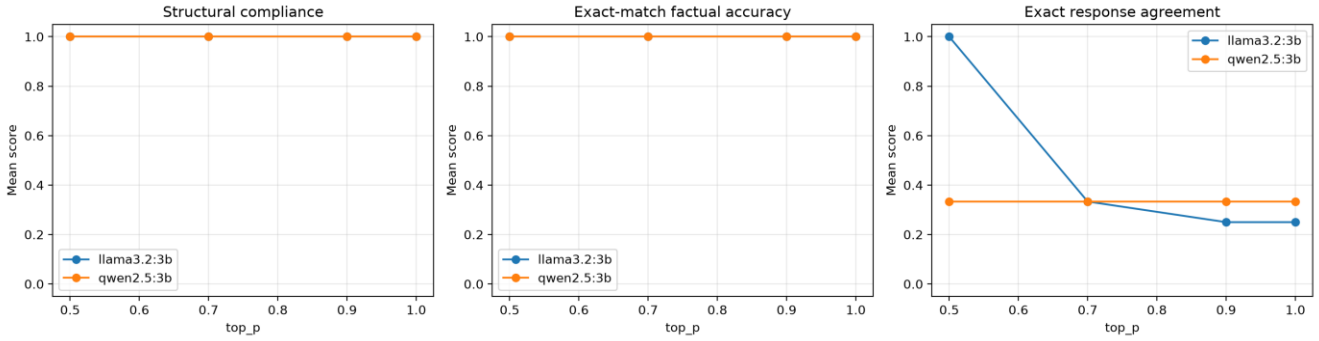
An agreement of 0% does not mean that all answers were wrong. It means that no pair of complete responses matched exactly. The observed change concerns repetition, not a measured decline in factual accuracy.

7. Top-K and Top-P



Above Fig. shows mean scores by Top-K, averaging over temperature and Top-P.

Top-K values of 10, 20, 50 and 100 produced the same averaged scores in this experiment. Llama maintained 45.83% exact agreement and Qwen maintained 33.33%. This gives no evidence of a Top-K effect on the reported averages for this prompt; it does not establish that Top-K has no effect on other tasks.



This shows mean scores by Top-P, averaging over temperature and Top-K.

Top-P	Llama agreement	Qwen agreement
0.5	100%	33.33%
0.7	33.33%	33.33%
0.9	25%	33.33%
1.0	25%	33.33%

Llama was most consistent at Top-P 0.5, with agreement decreasing at higher values. Qwen showed the same mean agreement at every Top-P value tested. Both models continued to score 100% on the short factual answers and structural check. The different patterns show that the two models did not respond identically to every sampling control.

8. Discussion and limitations

Temperature showed the clearest shared pattern: increasing it reduced exact repetition in both models. Top-P also affected Llama's consistency, while the reported Top-K averages remained unchanged. Because the plots average across other settings, the accompanying parameter summary should be used when comparing a particular combination.

The factual questions were easy enough that both models reached the maximum score. This limits what can be concluded about hallucination. No errors were detected in the three scored fields, but that finding is not evidence that the models were error-free across the whole response or would remain accurate on more demanding questions.

Only one prompt and three seeds per setting were used. Exact agreement treats punctuation, wording and JSON formatting differences as disagreement, even when meaning is unchanged. The experiment did not score creativity, usefulness, the accuracy of the seasons explanation, or its one-sentence requirement.

The submitted run did not record exact software versions, hardware details or immutable model identifiers. The model tags and generation settings are supplied, but exact text reproduction on another installation is not guaranteed. Results describe this experiment rather than a general ranking of the models.

9. Conclusion

For this prompt, temperature 0.0 gave the most repeatable responses in both models. Higher temperatures increased variation without reducing the measured short-answer accuracy or JSON compliance. Llama was more consistent overall, with mean exact agreement of 45.83%, compared with 33.33% for Qwen.

When identical responses are the priority, temperature 0.0 is supported by these results. For Llama, Top-P 0.5 also produced full agreement in this run. Higher temperatures can be considered when varied wording is wanted, but the experiment does not show whether that variation improves quality. A broader factual benchmark would be needed before recommending settings for hallucination control.